***LETTER OF REFERENCE***

# INDUSTRIAL CLIENT

To: THE ENGINEERING COMPANY

*March 16, 2026*

We would like to formally express our strong satisfaction with the successful implementation of the electrical energy saving and electrical system performance and power quality optimization project carried out by the company at the client's production facility.

From the initial engineering assessment through final commissioning, the company demonstrated a high level of technical expertise, organization, and commitment to delivering measurable results. A comprehensive measurement campaign was conducted across our electrical infrastructure, including motors, variable frequency drives (VFDs), transformers, and distribution subpanels. Advanced power quality instrumentation was used to record electrical parameters, harmonic distortion levels, and transient phenomena affecting our installation.

Based on the collected data, the company performed in-depth technical analysis and designed customized interventions tailored to the operational characteristics of our facility. Particular attention was given to improving overall system efficiency, reducing harmonic distortion, stabilizing voltage conditions, and enhancing the performance of critical production equipment.

**A key success factor of this project was the exceptionally fast implementation timeline.** The entire scope — from detailed measurements and engineering analysis to installation and full commissioning — was completed within a record timeframe.

This rapid execution was of critical importance to the client, as our production operates on a strictly seasonal basis, with peak industrial activity concentrated between July and September. Thanks to the company's efficiency and coordination, the project was fully operational in time to support a significant portion of our high-production September period. Delivering a technically complex energy optimization project within such a compressed schedule — while ensuring full functionality and reliability — was a major achievement and of substantial strategic value to our organization.

Following commissioning, the project's performance was evaluated through a comprehensive and technically rigorous approach.

Initially, detailed real-time comparative electrical measurements were conducted across the facility, confirming a substantial and immediate reduction in electrical consumption.

Beyond conventional measurement techniques, the company applied a highly advanced Artificial Intelligence–based energy performance modeling methodology. Using historical operational data — including production volumes, raw material usage, operating hours, and environmental conditions — the company developed a predictive digital energy model capable of accurately simulating our facility's electricity consumption under varying operational scenarios.

The model was trained using pre-installation data and validated to ensure high predictive accuracy. It was then applied to the post-installation operating period to determine the theoretical energy consumption that would have occurred without the implementation of the company solution.

By comparing the AI model's predicted consumption with the actual metered energy consumption after project implementation, **the verified total energy saving achieved across the entire facility reached: 11.89% of total electrical energy consumption.**

It is important to highlight that the guaranteed performance target for the project was **10.10%**, meaning that the achieved result significantly exceeded the initial commitment.

The consistency between physical electrical measurements and the Artificial Intelligence predictive analysis further reinforces the credibility, robustness, and scientific validity of the achieved result.

In addition to the significant energy savings, **the project delivered multiple qualitative improvements to our electrical infrastructure, including:**

- **Mitigation of voltage harmonics from 6.5% to 1.9% and current harmonics from 49% to 2.9%,**
- **Improved operational efficiency of motors, VFDs, and transformers,**
- **Increased electrical system stability through voltage stabilization at nominal levels and increased reserve capacity by 26%,**
- **Reduction of voltage-related disturbances affecting production continuity,**
- **Enhanced reliability during peak seasonal production loads.**

The combination of rapid execution, advanced engineering methodology, and measurable performance improvement clearly demonstrates the company's technological capability and professionalism.

Based on the overall performance of the project and the tangible technical and economic benefits delivered to the client, we consider this collaboration to be highly successful and would confidently recommend the company as a technologically advanced and reliable partner for complex energy efficiency and electrical system performance and power quality optimization projects.

Sincerely,

For the industrial client

Technical Director